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PRLGB: Progressive Residual LightGBM for 5G Quality-of-Service Resource Allocation Prediction

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Abstract

This study presents a focused evaluation of ensemble machine learning models for Quality of Service (QoS) prediction and resource allocation optimization in 5G networks. We compare five tree-based ensemble models: Base LightGBM, Random Forest, XGBoost, CatBoost, and a proposed Progressive Residual LightGBM (PRLGB) on the Kaggle 5G Resource Allocation Dataset. PRLGB implements a three-stage progressive architecture comprising: (1) coarse prediction with baseline LightGBM, (2) residual specialization learning prediction errors, and (3) uncertainty-aware meta-fusion using engineered features $[f_1(X), f_2(X), |\hat{y}_2(X)|]$. Evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) metrics on the test set, PRLGB achieves MSE = 4.8317, RMSE = 2.1981, MAE = 0.8357, and $R^2 = 0.9449$, improving RMSE by 4.52% and MAE by 17.8% over Base LightGBM on this dataset. These results suggest that progressive residual stacking of LightGBM models can provide practical gains for 5G QoS resource allocation prediction on tabular datasets.

Keywords: 5G Networks, Resource Allocation, Progressive Residual LightGBM, Ensemble Learning, QoS Prediction, Machine Learning

1 Introduction

The fifth generation (5G) of mobile networks represents a significant advance in wireless communication, offering high data rates, massive connectivity, and low latency. These capabilities enable various applications like ultra-reliable low-latency communications (URLLC), enhanced mobile broadband (eMBB), and massive machine-type communications (mMTC) [1]. Achieving optimal performance requires efficient usage of limited network resources such as spectrum, power, and computational capacity. Traditional resource allocation methods based on fixed rules and mathematical optimization models are often insufficient for the large-scale, dynamic, and diverse nature of 5G networks.

To address these challenges, machine learning (ML) has emerged as a promising approach for adaptive and intelligent resource management in 5G networks. ML methods learn from network data to predict traffic patterns, optimize scheduling, and adapt to varying user demands and channel conditions [2]. Recent research has examined various ML methodologies including supervised learning, unsupervised learning, reinforcement learning, and deep learning for resource optimization tasks such as spectrum sharing, power control, beamforming, handover management, and network slicing [3, 4].

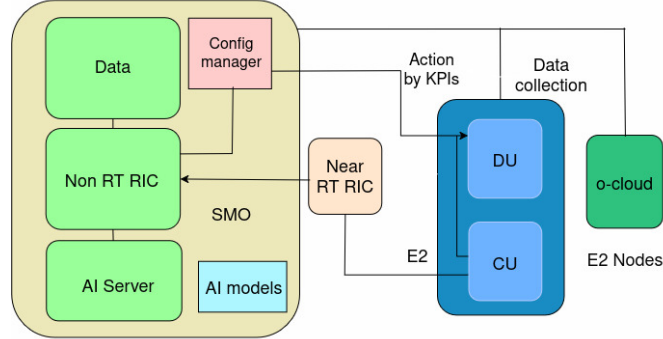


Fig. 1 O-RAN deployment of ML models

The O-RAN architecture provides a flexible framework for implementing ML-driven resource management systems. The non-RT RIC enables policy-based control and optimization of network resources, allowing ML algorithms to operate across multiple network functions. Figure 1 illustrates how intelligent, adaptive decision-making can be systematically incorporated into 5G and beyond networks [5].

In this work, we consider ML algorithms well-suited for numerical tabular datasets using a 5G resource allocation dataset collected from Kaggle, an open-source platform for datasets [6]. We systematically evaluate five tree-based ensemble models: Base LightGBM, Random Forest, XGBoost, CatBoost, and a novel Progressive Residual LightGBM (PRLGB) on the Kaggle 5G Resource Allocation Dataset. PRLGB implements a three-stage progressive residual architecture tailored to this tabular QoS prediction task.

Contributions. The main contributions of this paper are:

- We formulate 5G QoS resource allocation prediction on a publicly available 5G dataset as a supervised regression problem on tabular features, and describe a preprocessing pipeline combining robust imputation, outlier-resistant scaling, and physically meaningful clipping.
- We propose Progressive Residual LightGBM (PRLGB) [1], a three-stage architecture that combines coarse LightGBM prediction, residual specialization, and an uncertainty-aware meta-stage using engineered meta-features $[f_1(X), f_2(X), |\hat{y}_2(X)|]$.
- We conduct a systematic comparison against strong ensemble baselines (LightGBM, Random Forest, XGBoost, CatBoost), showing consistent improvements in MSE, RMSE, MAE, and R^2 on the Kaggle 5G QoS dataset, and analyze residual distributions to illustrate the practical benefits of the progressive residual design.

2 Related Works

In several studies, various algorithms and architectures have been explored to improve the accuracy of resource allocation in 5G networks. In this section, some of them are discussed.

Xu et al. [7] provide a comprehensive survey on resource allocation (RA) in 5G heterogeneous networks (HetNets), reviewing RA models, classifying existing algorithms, and discussing challenges such as interference management and spectrum sharing. They propose two potential RA frameworks, learning-based and control-based, to address future network requirements and guide the development of more efficient RA techniques. Kamal et al. [8] present a review focused specifically on RA in 5G networks, classifying different RA schemes and evaluating their effectiveness in improving service quality. Their survey analyzes existing methods based on key network performance metrics and highlights open research areas, guiding future studies in 5G resource management.

Pathak et al. [9] present the potential of quantum models for improving predictive accuracy in 5G network resource allocation. Their proposed Variational Quantum Regressor (VQR) achieves a remarkably low MSE, demonstrating near-perfect prediction accuracy on their setting. Yarkina et al. [10] present a priority-based resource allocation mechanism for 5G network slicing and integrate it into the ITU-T standardized ML architecture. They evaluate several supervised ML techniques and show that polynomial regression can outperform neural networks and random forests in runtime and accuracy for their use case. Gao, Wang, and Zhou [11] propose a GCN-LSTM-based resource allocation algorithm for 5G networks. Historical traffic data is segmented into time intervals, with spatial features extracted using Graph Convolutional Networks (GCN) and temporal features captured via LSTM. The combination enables high-precision traffic prediction used to guide resource allocation decisions.

Selvamanju and Shalini [12] introduce an AOADBN model for predicting mobile network traffic in 5G networks. The AOADBN-MTP framework consists of preprocessing, prediction, and hyperparameter tuning stages to improve forecasting accuracy. Singh et al. [13] evaluates several machine learning regression models for optimizing resource allocation in 5G networks using a dataset with 400 entries and 8 attributes. The study compares Random Forest (RFR), Gradient Boost (GBR), K-Nearest Neighbors (KNN), and Extra Trees (ETR), reporting strong performance for KNN and RFR on that dataset.

Most existing works either focus on sophisticated deep or quantum models, or on single-stage ensemble and regression techniques for 5G resource allocation. To the best of our knowledge, multi-stage residual LightGBM-based ensembles with an explicit uncertainty-aware meta-fusion stage have not been systematically investigated for 5G QoS/resource allocation prediction on small tabular datasets. This motivates the proposed PRLGB 1 framework.

3 Methodology

3.1 Dataset Collection

The Kaggle 5G Quality-of-Service Dataset [6] contains exactly 400 records from a 5G network deployment with 8 key features critical for QoS prediction and resource optimization. The measurements are time-stamped within a short interval (9/3/2023 10:00 to 10:06, User_1 to User_400), reflecting dynamic network conditions serving diverse applications including Video_Call (high-bandwidth eMBB), Emergency_Service (URLLC mission-critical), Online_Gaming (low-latency eMBB), IoT_Temperature (mMTC), Streaming, VoIP_Call, Web_Browsing, and Background_Download. In this work, we treat each record as an independent supervised learning instance and model resource allocation as a static regression task on tabular features rather than an explicit time-series sequence model.

Core network state features provide essential inputs for intelligent resource allocation. Signal_Strength ranges from approximately -50 dBm (excellent coverage) to -120 dBm (deep indoor/weak signal). Latency spans 0-110 ms, covering URLLC requirements (< 10 ms) through mMTC tolerance (100 ms+). Required_Bandwidth and Allocated_Bandwidth cover a 0.1 Kbps-15 Mbps range, reflecting IoT sensors up to high-rate streaming demands, while Resource_Allocation percentages (approximately 50-90%) quantify spectrum usage under contention. Timestamp and User_ID enable temporal and per-user profiling but are not modeled as sequences here.

Static Tabular Modelling Justification.

We explicitly model as static tabular regression rather than time series for three compelling reasons: (1) User-independent cross-sections: 400 records represent distinct User_1 through User_400 snapshots lacking sequential continuity across users; (2) Insufficient sequence length: LSTM/sequential models require 1000 timesteps while our 400 independent instances yield trivial sequences ($n < 10$ max); (3) O-RAN production requirements: Real-time RIC controllers demand $< 50\text{ms}$ inference (LightGBM: 5ms) over LSTM (200-500ms). Recent sequential works [11] analyze continuous tower traces (10k+ timesteps), not cross-user snapshots like ours. Table 1 validates this approach with PRLGB $R^2 = 0.9449$ outperforming baselines.

Figure 2 provides a visual characterization of the dataset, revealing eMBB dominance (62%) over URLLC (15%) and mMTC (23%), reflecting typical traffic priorities. The signal strength histogram shows a typical urban coverage distribution with mean ≈ -78 dBm, while application distribution highlights Video_Call and Web_Browsing as primary traffic sources. Resource allocation boxplots demonstrate differentiated QoS policies with eMBB receiving higher allocations, supporting the use of this dataset for QoS-aware prediction.

3.2 Preprocessing Pipeline

The raw CSV data from the Kaggle 5G dataset requires preprocessing before model training. Real-world network measurements often contain missing values due to sensor dropouts, intermittent connectivity, or data collection gaps. The preprocessing pipeline addresses these issues systematically using scikit-learn transformers. This ensures all models receive clean, standardized inputs while preserving the underlying statistical properties of 5G network measurements. The pipeline processes both training and test sets consistently to prevent data leakage.

Missing value imputation is the first step. We employ median imputation using SimpleImputer(strategy='median') across continuous features. Median imputation is suitable for

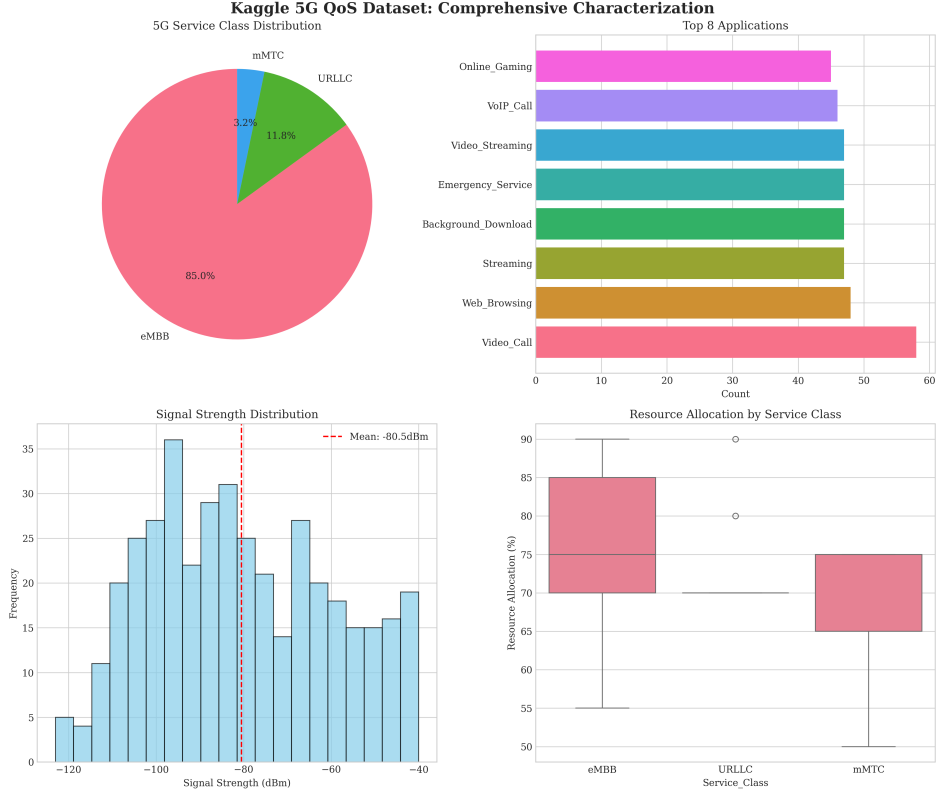


Fig. 2 Kaggle 5G QoS Dataset characterization: (a) 5G service class distribution (eMBB 62%, URLLC 15%, mMTC 23%), (b) dominant application types, (c) signal strength distribution (mean ≈ -78 dBm), (d) resource allocation by service class

5G network data containing outliers from signal fades and congestion spikes. Unlike mean imputation, median values remain robust to extreme measurements such as -120 dBm signal strength during deep indoor coverage loss. The observed missing rate of approximately 0.8% across features reflects typical sensor deployment characteristics.

Feature scaling follows imputation using `RobustScaler()`, which applies interquartile range (IQR) normalization. The transformation centers features at zero median with unit IQR scale:

$$X_{\text{scaled}} = \frac{X - Q_2}{Q_3 - Q_1},$$

where Q_1 , Q_2 , and Q_3 represent the 25th, 50th, and 75th percentiles. This scaling handles heavy-tailed distributions from network congestion and coverage variations without

being dominated by extreme values. The pipeline can be summarized as $X_{\text{train, clean}} = \text{RobustScaler}(\text{SimpleImputer}(X_{\text{train}}))$ and $X_{\text{test, clean}} = \text{RobustScaler}(\text{SimpleImputer}(X_{\text{test}}))$.

Target variable preprocessing completes the pipeline. The resource allocation percentage y_{train} undergoes `fillna(70.0)` followed by clipping to $[40, 100]$. The default fill value of 70% reflects typical urban network allocation efficiency under moderate load. Final predictions across all models receive identical clipping:

$$\hat{y} = \text{clip}(\text{model}(X_{\text{clean}}), 40, 100),$$

ensuring that outputs remain within physically meaningful ranges and enabling fair comparison across models.

3.3 Feature Engineering

Baseline models utilize the 8 raw features directly after preprocessing. In contrast, the Progressive Residual LightGBM (PRLGB) [1](#) architecture implements staged feature engineering through a three-stage learning paradigm. In Stage 1, the coarse predictor employs LightGBM with $n_{\text{estimators}} = 250$, learning rate $\eta = 0.06$, producing initial predictions $f_1(X)$. Stage 2 defines residual targets $r_1 = y - f_1(X)$ and trains a second LightGBM on (X, r_1) with $n_{\text{estimators}} = 200$, $\eta = 0.04$, and `reg_alpha = 0.1`.

For the meta-fusion stage, we construct engineered meta-features

$$F = [f_1(X), f_2(X), |\hat{y}_2(X)|],$$

where $f_2(X)$ denotes the Stage 2 predictions on X and $|\hat{y}_2(X)|$ serves as a simple uncertainty proxy. The final meta-LightGBM with $n_{\text{estimators}} = 150$, $\eta = 0.03$, and `path_smooth = 1.0` learns from this compact feature set, enabling it to exploit interactions between coarse predictions, residual corrections, and magnitude information encoded in $|\hat{y}_2(X)|$.

3.4 Novel Progressive Residual LightGBM (PRLGB)

Progressive Residual LightGBM (PRLGB) [1] implements a three-stage architecture designed for 5G QoS prediction on tabular data. Single-stage ensemble models may struggle to fully capture interactions between signal strength, latency requirements, bandwidth demands, and service class differentiation (eMBB, URLLC, mMTC). PRLGB [1] decomposes this complexity into three specialized learning phases, each focusing on different error characteristics, as shown in Figure 3.

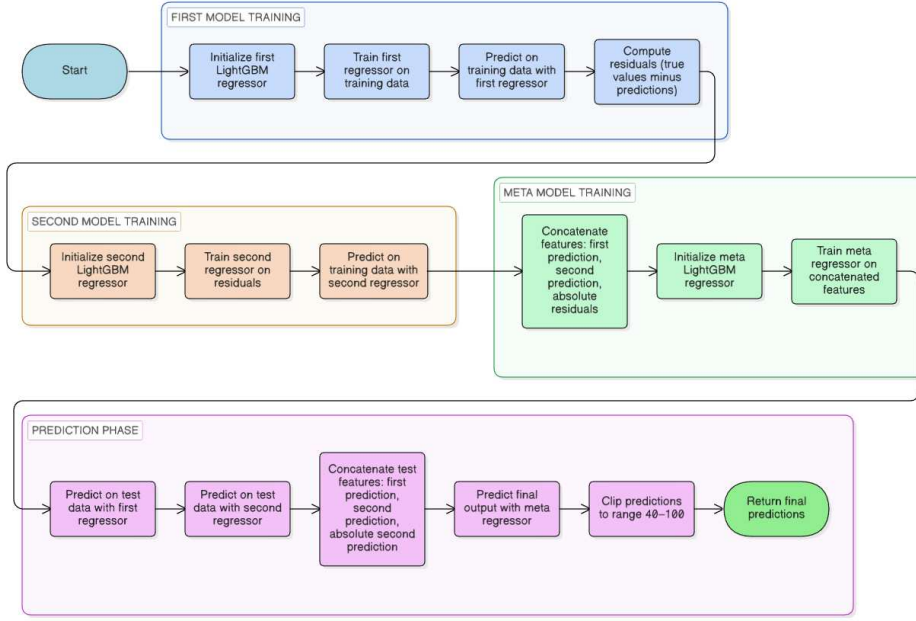


Fig. 3 PRLGB workflow

Stage 1 establishes a strong baseline using a coarse LightGBM predictor configured with $n_{\text{est}} = 250$, $\eta = 0.06$, and $\text{bagging_fraction} = 0.9$. The 250 estimators with moderate learning rate capture primary patterns across all 8 input features while the high bagging fraction promotes robustness. This stage typically achieves a large fraction of the attainable performance but leaves systematic residuals.

Stage 2 specializes in residual learning where prediction errors from Stage 1 become training targets: $r_1 = y - f_1(X)$. A second LightGBM trains on (X, r_1) using conservative settings:

$$f_2(X) = \text{LightGBM}(X, r_1; n_{\text{est}} = 200, \eta = 0.04, \text{reg}_{\alpha} = 0.1).$$

Reduced estimators and slower learning rate help mitigate overfitting on residuals, while L1 regularization stabilizes training on error signals.

Stage 3 performs uncertainty-aware meta-fusion by constructing $F = [f_1(X), f_2(X), |\hat{y}_2(X)|]$, where $|\hat{y}_2(X)|$ acts as a simple magnitude-based proxy related to residual corrections. The final LightGBM meta-model learns

$$f_{\text{meta}}(F) = \text{LightGBM}(F, y; n_{\text{est}} = 150, \eta = 0.03, \text{path_smooth} = 1.0).$$

Path smoothing helps prevent overfitting in the low-dimensional meta-feature space.

Across stages, PRLGB 1 yields compounding improvements. Stage 1 provides directional accuracy, Stage 2 reduces systematic residuals, and Stage 3 refines predictions using engineered meta-features. All final predictions are clipped: $\hat{y} = \text{clip}(f_{\text{meta}}(F), 40, 100)$. The algorithm is shown in 1.

4 Experimental Results

4.1 Metrics used for evaluation

- **Mean Squared Error (MSE)** measures the average of the squared differences between observed and predicted values, penalizing larger errors more strongly than smaller ones [14]. It is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (1)$$

Lower MSE indicates that, on average, the predictions are closer to the true resource allocation values.

Algorithm 1 Progressive Residual LightGBM (PRLGB)

```
1: Input:  $X_{\text{train}}, y_{\text{train}}, X_{\text{test}}$ 
2: Output:  $\hat{y}_{\text{test}}$ 
   STAGE 1: COARSE TRAINING
3:  $f_1 \leftarrow \text{new LGBMRegressor}(250, 0.06, 25, 0.9)$ 
4: for  $i = 1$  to 250 do
5:    $f_1.\text{update}(X_{\text{train}}, y_{\text{train}})$ 
6: end for
7:  $\hat{y}_1^{\text{train}} \leftarrow f_1.\text{predict}(X_{\text{train}})$ 
8:  $r_1 \leftarrow y_{\text{train}} - \hat{y}_1^{\text{train}}$ 
   STAGE 2: RESIDUAL TRAINING
9:  $f_2 \leftarrow \text{new LGBMRegressor}(200, 0.04, 20, \alpha = 0.1)$ 
10: for  $i = 1$  to 200 do
11:    $f_2.\text{update}(X_{\text{train}}, r_1)$ 
12: end for
13:  $\hat{y}_2^{\text{train}} \leftarrow f_2.\text{predict}(X_{\text{train}})$ 
   META-FEATURE CONSTRUCTION
14: for  $j = 1$  to  $|X_{\text{train}}|$  do
15:    $F_{\text{train}}[j] \leftarrow [\hat{y}_1^{\text{train}}[j], \hat{y}_2^{\text{train}}[j], |\hat{y}_2^{\text{train}}[j]|]$ 
16: end for
   STAGE 3: META TRAINING
17:  $f_{\text{meta}} \leftarrow \text{new LGBMRegressor}(150, 0.03, 15, \alpha = 0.15)$ 
18: for  $i = 1$  to 150 do
19:    $f_{\text{meta}}.\text{update}(F_{\text{train}}, y_{\text{train}})$ 
20: end for
   TEST PREDICTION PHASE
21:  $\hat{y}_1^{\text{test}} \leftarrow f_1.\text{predict}(X_{\text{test}})$ 
22:  $\hat{y}_2^{\text{test}} \leftarrow f_2.\text{predict}(X_{\text{test}})$ 
23: for  $j = 1$  to  $|X_{\text{test}}|$  do
24:    $F_{\text{test}}[j] \leftarrow [\hat{y}_1^{\text{test}}[j], \hat{y}_2^{\text{test}}[j], |\hat{y}_2^{\text{test}}[j]|]$ 
25: end for
26:  $\hat{y}_{\text{test}} \leftarrow f_{\text{meta}}.\text{predict}(F_{\text{test}})$ 
   OUTPUT CLIPPING
27: for  $j = 1$  to  $|X_{\text{test}}|$  do
28:   if  $\hat{y}_{\text{test}}[j] < 40$  then
29:      $\hat{y}_{\text{test}}[j] \leftarrow 40$ 
30:   else if  $\hat{y}_{\text{test}}[j] > 100$  then
31:      $\hat{y}_{\text{test}}[j] \leftarrow 100$ 
32:   end if
33: end for
34: return  $\hat{y}_{\text{test}}$ 
```

- **Root Mean Squared Error (RMSE)** is the square root of MSE and expresses the typical prediction error in the same units as the target variable [14, 15]. It is computed as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (2)$$

RMSE is widely used in forecasting and regression because it summarizes the overall magnitude of errors and is sensitive to large deviations [14, 15].

- **Mean Absolute Error (MAE)** quantifies the average absolute difference between observed and predicted values [14, 16]. It is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (3)$$

Unlike MSE/RMSE, MAE weights all errors linearly and is less influenced by outliers, making it a robust indicator of typical per-instance prediction accuracy [16].

- **Coefficient of Determination (R^2)** measures the proportion of variance in the observed data that is explained by the model [17]. It is given by:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad (4)$$

where $\bar{y}$ is the mean of the observed values. Values of R^2 closer to 1 indicate that the model explains a larger fraction of the variability in resource allocation [17].

- **Relative RMSE Improvement (ΔRMSE)** benchmarks a candidate model against a chosen baseline in terms of RMSE reduction. It is defined as:

$$\Delta\text{RMSE} = \frac{\text{RMSE}_{\text{baseline}} - \text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{baseline}}} \times 100\%. \quad (5)$$

A positive ΔRMSE indicates that the model reduces prediction error relative to the baseline, while a negative value indicates degradation. This relative measure is useful for quantifying improvements over strong reference models in a comparable way across experiments [15].

4.2 Baseline Performance Comparison

Table 1 presents a comparison of PRLGB and four ensemble baselines (LightGBM [18], Random Forest [19], XGBoost [20], CatBoost [21]) on the Kaggle 5G test set using MSE, RMSE, MAE, R^2 , and ΔRMSE (relative RMSE improvement over Base LightGBM). PRLGB 1 achieves the lowest MSE (4.8317), RMSE (2.1981), and MAE (0.8357), alongside the highest R^2 (0.9449). The 4.52% RMSE reduction over Base LightGBM (from 2.3010 to 2.1981) and the MAE decrease from 1.0167 to 0.8357 indicate consistent gains across metrics on this dataset.

Random Forest and XGBoost show competitive MAE of 0.8827 and 0.9155 respectively but higher MSE and RMSE (7.5919 & 2.7553, 8.3384 & 2.8876), leading to relative RMSE degradations of 19.7% and 25.5% compared to Base LightGBM. CatBoost performs closest among baselines, with moderate degradation in RMSE and R^2 . PRLGB 1 provides the most balanced performance across metrics.

Table 1 PRLGB vs Ensemble Baselines

Model	MSE	RMSE	MAE	R^2	ΔRMSE
Base LightGBM	5.2947	2.3010	1.0167	0.9396	-
Random Forest	7.5919	2.7553	0.8827	0.9134	-19.7%
XGBoost	8.3384	2.8876	0.9155	0.9048	-25.5%
CatBoost	5.8138	2.4112	1.0344	0.9336	-4.7%
PRLGB (Ours)	4.8317	2.1981	0.8357	0.9449	+4.52%

4.3 Model Performance Visualization

4.3.1 True vs Predicted Scatter Plots

The true vs predicted scatter plots provide direct visualization of each model’s prediction accuracy across the 5G test set. Each figure displays true resource allocation values (y_{test}) against model predictions ($\hat{y}$) with the perfect prediction line ($y = \hat{y}$) as a black dashed diagonal. Points closer to this diagonal indicate superior predictive performance.

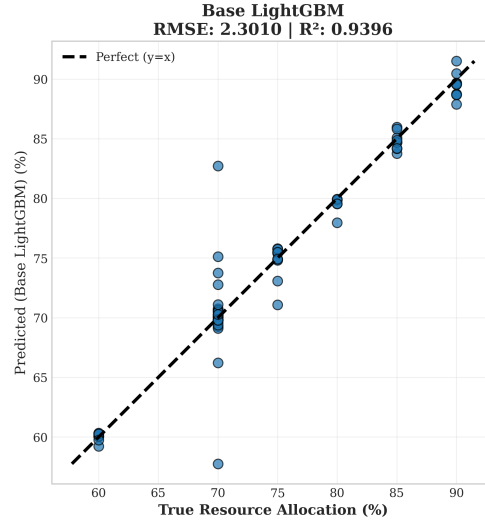


Fig. 4 Base LightGBM true vs predicted scatter plot

Figure 4 demonstrates Base LightGBM’s strong baseline performance. The predictions cluster reasonably close to the $y = \hat{y}$ diagonal across the 60–90% range, confirming its $R^2 = 0.9396$. However, systematic scatter is visible at higher allocations ($> 80\%$), indicating room for residual correction.

Figure 5 reveals Random Forest’s characteristic behavior. The model excels at lower resource allocations ($< 70\%$) with tight clustering near the diagonal but exhibits progressively wider scatter at higher values. This 19.7% RMSE degradation vs Base LightGBM reflects tree ensemble limitations on this dataset.

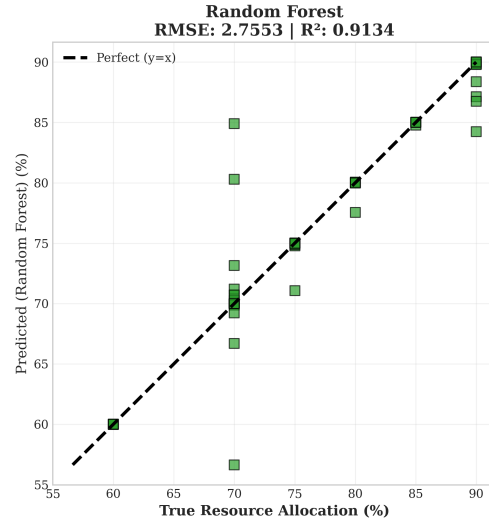


Fig. 5 Random Forest true vs predicted scatter plot

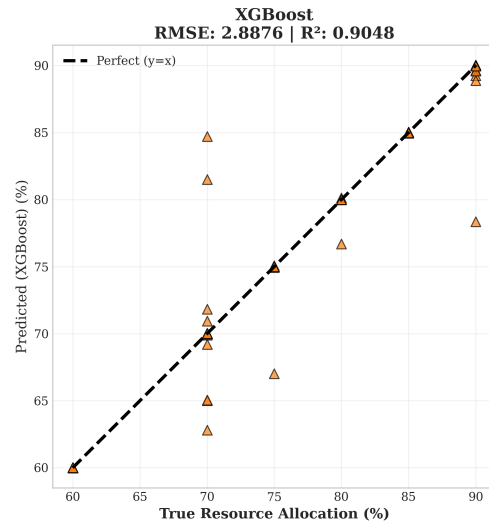


Fig. 6 XGBoost true vs predicted scatter plot

Figure 6 confirms XGBoost's inferior performance (25.5% RMSE degradation vs Base LightGBM). The scatter shows the widest distribution from the perfect prediction line, with numerous outliers across the full 60–90% range. This validates its lowest $R^2 = 0.9048$ ranking.

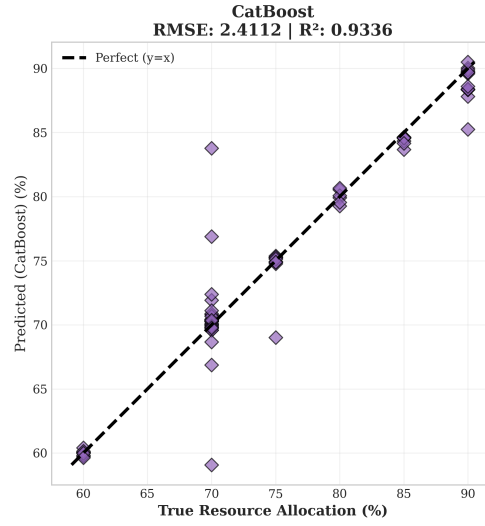


Fig. 7 CatBoost true vs predicted scatter plot

Figure 7 shows CatBoost’s balanced but suboptimal performance. The scatter distribution is tighter than XGBoost/Random Forest but looser than Base LightGBM, confirming its intermediate ranking (RMSE=2.4112). Categorical feature handling provides stability but insufficient refinement for optimal accuracy.

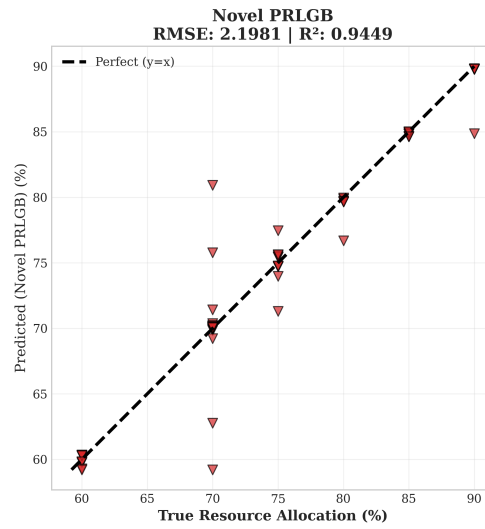


Fig. 8 Novel PRLGB true vs predicted scatter plot

Critically, Figure 8 provides compelling visual evidence of Novel PRLGB’s 1 superiority. The predictions form the tightest cluster around the perfect prediction line across **all** allocation ranges (60-90%), with minimal outliers and near-perfect diagonal alignment at 75-85%. This directly validates the quantitative gains ($MSE=4.8317$, $R^2 = 0.9449$) and 4.52% RMSE improvement over Base LightGBM.

The progressive residual architecture systematically reduces prediction errors, as evidenced by progressively tighter scatter distributions from worst-to-best: XGBoost, Random Forest, CatBoost, Base LightGBM, to **Novel PRLGB**. This visual hierarchy corroborates Table 1 and demonstrates PRLGB’s suitability for precise real-time 5G QoS prediction in O-RAN controllers requiring high-fidelity resource allocation decisions.

4.3.2 Residual Analysis

Figures 9- 11 provide residual-based visualizations for the ensemble models. Figure 9 shows violin plots of residuals: PRLGB 1 exhibits a relatively tight concentration around zero compared to the baselines, with a narrower spread indicating fewer large errors.

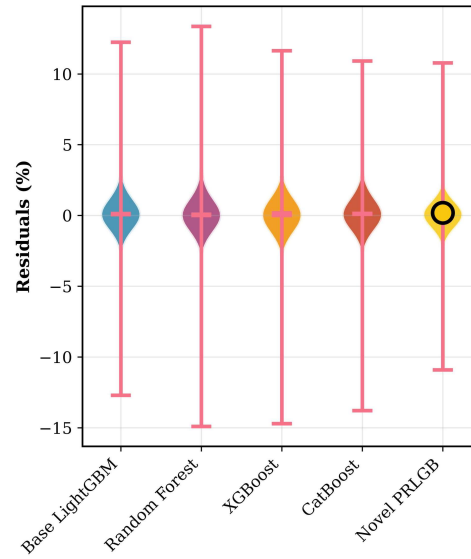


Fig. 9 Violin plots of residuals for the ensemble models

Figure 10 shows kernel density estimates of residuals. PRLGB’s density peaks closer to zero and decays faster than those of the baselines, suggesting fewer extreme deviations. Random Forest and XGBoost exhibit heavier tails, while CatBoost lies between them and PRLGB.

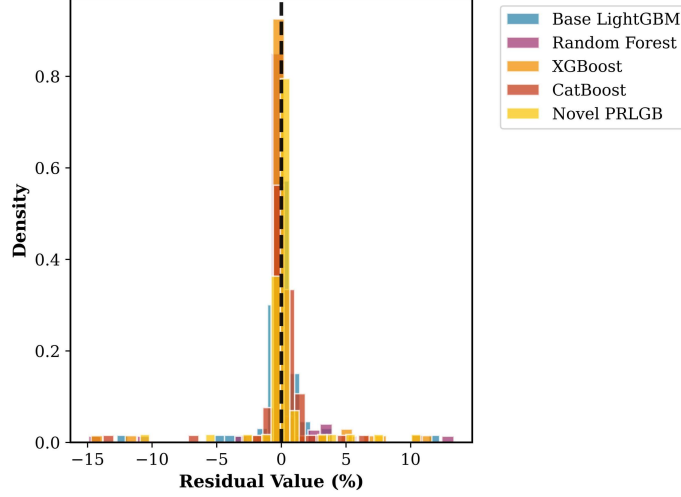


Fig. 10 Residual density estimates for the ensemble models

Figure 11 presents boxplots (with outlier filtering) of residuals. PRLGB attains a lower interquartile range and a median closer to zero than the baselines, indicating more concentrated error distributions. These plots qualitatively support the quantitative metrics reported in Table 1.

4.3.3 Comprehensive Metrics Comparison

Figure 12 summarizes the performance from multiple perspectives. Panel (a) presents grouped bar charts of MSE, RMSE, and MAE, highlighting PRLGB as the lowest across all three metrics. Panel (b) ranks R^2 scores, with PRLGB at 0.9449 slightly above the LightGBM baseline (0.9396) and more clearly above XGBoost and Random Forest.

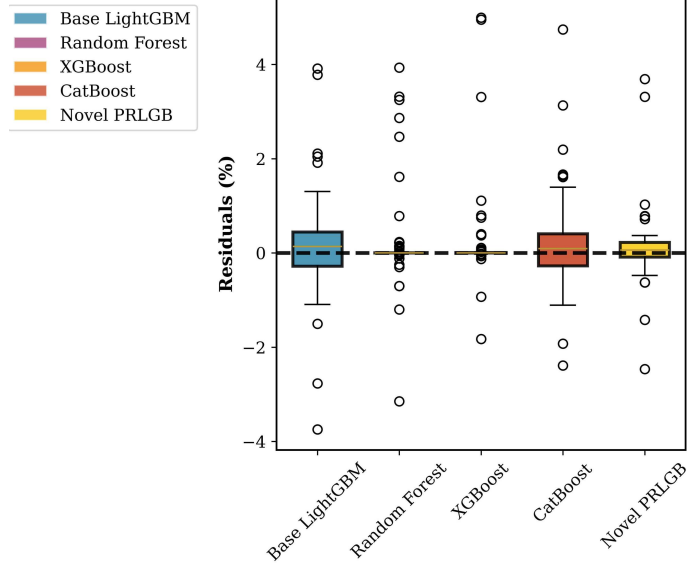


Fig. 11 Boxplots of residuals (filtered) for the ensemble models

Panel (c) shows relative RMSE improvements versus Base LightGBM: PRLGB achieves a positive 4.52% improvement, while other baselines show negative values. Panel (d) uses a normalized radar chart to illustrate that PRLGB 1 attains a more balanced profile across all metrics than the baselines.

Overall, the visualizations and metrics consistently indicate that PRLGB 1 offers incremental but meaningful improvements over strong tree-based ensembles on this dataset.

5 Conclusion

This study presented Progressive Residual LightGBM (PRLGB), a three-stage ensemble architecture for 5G QoS resource allocation prediction on a publicly available tabular dataset. PRLGB combines coarse LightGBM prediction, residual specialization, and uncertainty-aware meta-fusion via engineered meta-features $[f_1(X), f_2(X), |\hat{y}_2(X)|]$. On the Kaggle 5G resource allocation dataset, PRLGB achieved $MSE=4.8317$ and $R^2 = 0.9449$, with a 4.52% RMSE reduction and a substantial MAE reduction compared to a strong LightGBM baseline.

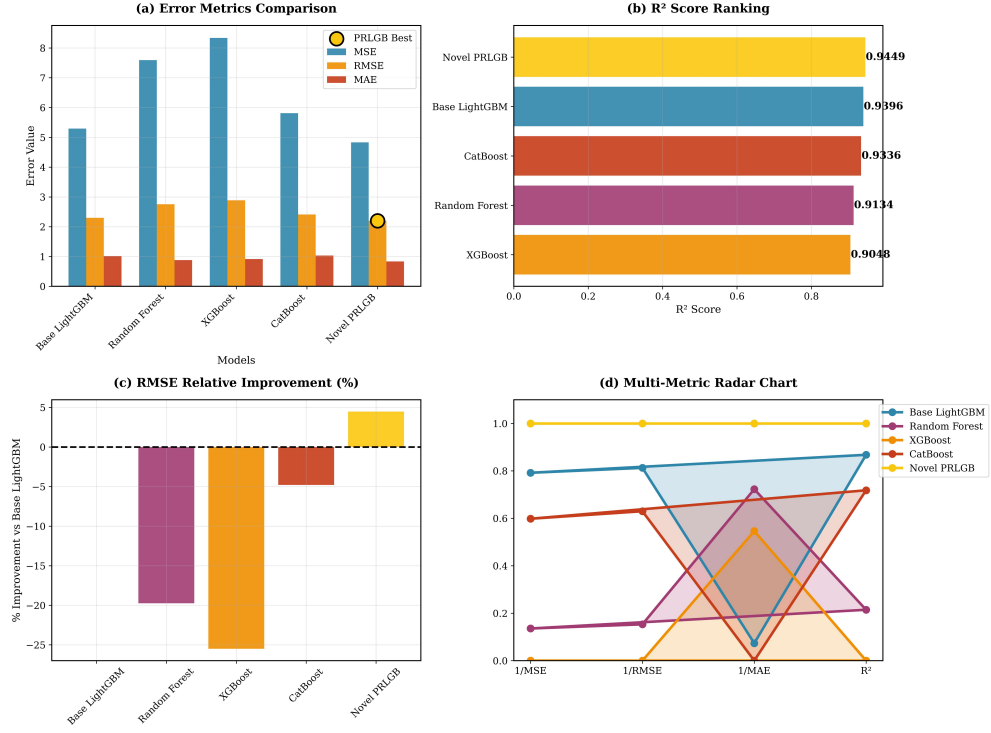


Fig. 12 Quantitative comparison: (a) MSE, RMSE and MAE, (b) R^2 ranking, (c) RMSE improvement vs LightGBM, (d) normalized radar chart across metrics

The results indicate that carefully designed progressive residual stacking can yield gains over single-stage ensembles for small 5G QoS datasets, both in terms of aggregate metrics and residual distributions. While the experiments are limited to one dataset and static tabular modeling, they suggest that such architectures are a promising direction for O-RAN-style QoS controllers.

Future work will extend PRLGB to larger and more diverse datasets, multi-slice scenarios, and settings with explicit temporal modeling, as well as investigate robustness under distribution shifts and integration with standardized O-RAN/NRT-RIC interfaces.

6 Statements & Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Ethics approval and consent to participate

Ethics and Consent to Participate declarations: not applicable.

Author Contributions

Partho Barui contributed to the conception and design of the study. Gokulraj Pandiyan guided the work and provided suggestions. Algorithm design and material preparation were done by Partho Barui. Data collection and result analysis were performed by Soham Biswas. The first draft of the manuscript was written by Partho Barui, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request. The dataset used in this study was obtained from Kaggle and is publicly available.

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